Modular flow generates generalized entropy in type-II_∞ crossed-product gravity: an Araki-cocycle derivation, with a no-go for HPS–DEHK Stinespring intertwining

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Abstract

Working in the type-II_∞ crossed-product algebra $\mathcal{M}_R = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$ attached to a static-patch observer (Chandrasekaran–Longo–Penington–Witten, de Vuyst–Eccles–Höhn–Kirklin), we use the Araki relative modular operator $\Delta_{\rho|\sigma}$ and its cocycle identity to derive the algebraic identity

$$\frac{d}{d\tau_R} S_{\mathrm{gen}}(\rho)\Big|_{\rho \text{ fixed},\, \sigma_{\tau_R}} = \langle K_R \rangle_\rho,$$

where K_R is the dressed modular Hamiltonian and $\sigma_{\tau_R} = \sigma \circ \sigma^\sigma_{-\tau_R}$ is the modular-time-evolved KMS reference state. The derivation is rigorous in the type-II_∞ setting via Pusz–Woronowicz applied to normal semifinite faithful (nsf) weights and is independent of the Petz recovery construction (which has no known type-II_∞ realization for $\mathcal{M}_R$).

We then examine, and refute at the operator level, the natural attempt to intertwine K_R with the chord (transfer-matrix) Hamiltonian H_{grav} of the double-scaled SYK / sine-dilaton gravity model (Heller–Papalini–Schuhmann; Heller–Ori–Papalini–Schuhmann–Wang) through a modular-rigidified Stinespring isometry $V : \mathcal{H}_{\mathrm{HPS}} \to \mathcal{H}_{\mathrm{DEHK}}$. The HPS chord spectrum $E(\theta) = -\cos\theta/\sqrt{2|\log q|(1-q^2)}$ is bounded continuous on $[-E_{\max}, +E_{\max}]$, while $K_R = \beta_{\mathrm{dS}} H_\xi + p$ is unbounded continuous on $\mathbb{R}$. No isometry can intertwine a bounded self-adjoint operator with an unbounded one; hence the proposed bridge fails as an operator-level identification.

What does survive is a thermodynamic / Krylov-asymptotic match: the semiclassical large-N partition functions agree, $\log Z_{\mathrm{chord}}(\beta_{\mathrm{dS}}) = \log Z_{\mathrm{DEHK}}(\beta_{\mathrm{dS}}) + O(1/N)$, reproducing the standard Bekenstein–Hawking de Sitter entropy $A_{\mathrm{horizon}}/(4G_N)$. We frame this paper as a *consistency note* (MVT2-class): the Araki theorem stands as a new result of independent interest, while the Stinespring bridge between the DSSYK chord algebra and the de Sitter crossed product is recorded as a no-go obstruction for future work.

1 Introduction

The modular flow / generalized entropy programme. Algebraic methods have transformed the discussion of generalized entropy in gravitational subregions. After Witten's observation [1] that gravitational backreaction promotes the type-III_1 algebra of bulk quantum field theory observables to a type-II crossed product, two parallel programmes have emerged: (i) the de Sitter / static-patch programme of Chandrasekaran–Longo–Penington–Witten [2] (CLPW), refined by de Vuyst–Eccles–Höhn–Kirklin [3] (DEHK), in which the crossed-product trace yields finite observer-dependent entropies, and (ii) the generalized second law (GSL) programme of Faulkner–Speranza [4] (FS24), in which monotonicity of relative entropy under modular inclusions translates to GSL-type statements for arbitrary cuts of Killing horizons.

Both programmes share a central structural object: the *modular Hamiltonian* K_R of the dressed (crossed-product) algebra. In the KMS-thermal de Sitter setting at $\beta_{\mathrm{dS}} = 2\pi r_{\mathrm{dS}}$ (Bisognano–Wichmann [14]; Connes–Rovelli thermal-time hypothesis [15]), K_R generates a genuine one-parameter automorphism group, and its expectation value $\langle K_R \rangle_\rho$ in a perturbed state ρ controls the first-law variation of the dressed entropy.

Two questions this note answers.

1. Does the Araki relative modular operator $\Delta_{\rho|\sigma}$ [7] provide a clean type-II_∞ derivation of $dS_{\mathrm{gen}}/d\tau_R = \langle K_R \rangle_\rho$, without invoking Petz recovery? *Yes* (Theorem 1).

2. Can the chord Hamiltonian H_{grav} of the DSSYK / sine-dilaton holographic dictionary of Heller–Papalini–Schuhmann [5] (HPS), refined by Heller–Ori–Papalini–Schuhmann–Wang [6] (HOPSW), be intertwined with K_R via a CPTP isometry between the chord Hilbert space $\mathcal{H}_{\mathrm{HPS}}$ and the GNS Hilbert space $\mathcal{H}_{\mathrm{DEHK}}$? *No* (Proposition 1), because their spectra have incompatible support.

The negative answer is sharper than a generic "algebra-type-mismatch" argument: it pinpoints the obstruction in a concrete, falsifiable way (bounded versus unbounded continuous spectrum). The thermodynamic match that survives, $\log Z_{\mathrm{chord}} = \log Z_{\mathrm{DEHK}} + O(1/N)$, is *not* new (it reduces to the Bekenstein–Hawking entropy of the cosmological horizon), and we therefore frame this paper at the level of a consistency note: a clean lemma (Theorem 1) plus a clean obstruction (Proposition 1) plus an honest account of what is left open.

Plan. Section 2 states and proves the Araki-cocycle identity $dS_{\mathrm{gen}}/d\tau_R = \langle K_R \rangle_\rho$ in type-II_∞. Section 3 reviews the modular-rigidified Stinespring embedding "C_{BS}" of the chord Hilbert space into the GNS Hilbert space of $\mathcal{M}_R$, and isolates the operator-level requirements (P1)–(P3) that any genuine bridge would have to satisfy. Section 4 states the spectrum-mismatch obstruction. Section 5 records the Krylov-thermodynamic match that does survive. Section 6 discusses alternative routes (Faulkner–Hollands II type-changing recovery, modular Krylov density-of-states matches) and open problems.

2 Wk2: Araki cocycle in type-II$_\infty$

2.1 Setup

Let $\mathcal{M}$ be a von Neumann algebra in standard form on a Hilbert space $\mathcal{H}$, and let σ be a faithful normal state with GNS vector $|\Omega_\sigma\rangle$ in the natural positive cone $\mathcal{P}^\sharp$. For a second faithful normal state ρ with GNS vector $|\Omega_\rho\rangle \in \mathcal{P}^\sharp$, the *Araki relative modular operator* [7] is

$$\Delta_{\rho|\sigma} \;=\; S^*_{\rho|\sigma}\, S_{\rho|\sigma}, \qquad S_{\rho|\sigma} : A|\Omega_\sigma\rangle \longmapsto A^*|\Omega_\rho\rangle, \quad A \in \mathcal{M}. \tag{1}$$

Let σ_t^σ denote the modular automorphism group of σ on $\mathcal{M}$, generated by the modular Hamiltonian $K_R = -\log \Delta_\sigma \geq 0$ (uniform sign convention, cf. Wk2 memo [20]). The *Araki relative entropy* is

$$S(\rho\|\sigma) \;=\; -\,\langle\Omega_\rho|\, \log \Delta_{\rho|\sigma}\, |\Omega_\rho\rangle, \tag{2}$$

finite when $\mathrm{supp}(\rho) \subset \mathrm{supp}(\sigma)$ and $+\infty$ otherwise.

2.2 Type-II$_\infty$ extension via Pusz–Woronowicz

For the de Sitter crossed product $\mathcal{M}_R = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$ of CLPW [2] and DEHK [3], $\mathcal{M}_R$ is type-II$_\infty$ and admits a normal semifinite faithful (nsf) tracial weight τ_{II} (the Haagerup trace). The Araki construction is purely algebraic and the Tomita–Takesaki theory applies to any standard form, so $\Delta_{\rho|\sigma}$ is well-defined for every pair of nsf weights. The Pusz–Woronowicz [8] variational characterization

$$S(\rho\|\sigma) \;=\; \sup_{\substack{A\in\mathcal{M}\\ A=A^*\geq 0}} \big\{\, \rho(\log A) - \log \sigma(e^A) \,\big\} \tag{3}$$

extends to type-II$_\infty$ weights through Haagerup's L^p theory and the Ohya–Petz [10] formulation in semifinite algebras.

2.3 The cocycle identity and modular-time derivative

The relative modular operator satisfies the Connes-cocycle identity (a direct consequence of σ_t^σ acting on the GNS data):

$$\Delta_{\rho|\sigma\circ\sigma^\sigma_{-t}} \;=\; \Delta_{\rho|\sigma}\, e^{tK_R}, \tag{4}$$

in the strong-resolvent sense on the joint domain. (See Bratteli–Robinson [9], Theorem 5.3.10 for the analyticity strip $0 \leq \mathrm{Im}\, t \leq \beta/2$ of Δ_σ^{it} on KMS states, and Theorem 5.3.15 for the differentiability of relative entropy along the modular flow.)

Theorem 1 (Araki rate, Wk2)**.** *Let $\mathcal{M}_R$ be a type-II$_\infty$ crossed-product algebra with nsf weight τ_{II}, KMS reference state σ at inverse temperature β_{dS} with modular Hamiltonian K_R, and let ρ be a normal state with* $\mathrm{supp}(\rho) \subset \mathrm{supp}(\sigma)$ *satisfying the regularity condition $\rho \in \mathrm{dom}\, K_R$ (i.e. $\langle K_R\rangle_\rho < \infty$). Define the modular-time-evolved reference $\sigma_{\tau_R} = \sigma \circ \sigma^\sigma_{-\tau_R}$. Then*

$$\frac{d}{d\tau_R}\, S\big(\rho\|\sigma_{\tau_R}\big)\Big|_{\tau_R=0} \;=\; -\,\langle K_R\rangle_\rho, \tag{5}$$

and consequently, with the FS24 identification $S_{\rm gen}(\rho) = S_{\rm gen}(\sigma) - S(\rho\|\sigma) +$ (KMS boundary terms) [4],

$$\boxed{\left.\frac{dS_{\rm gen}}{d\tau_R}\right|_{\rho\ \textit{fixed}} \;=\; +\langle K_R\rangle_\rho.} \tag{6}$$

Proof. By (4), $\log\Delta_{\rho|\sigma_{\tau_R}} = \log\Delta_{\rho|\sigma} + \tau_R K_R$ on the joint analyticity domain. Differentiating the formula (2), exchanging the $d/d\tau_R$ with the GNS expectation (justified by the dominated convergence on the strip via the KMS analyticity of Bratteli–Robinson [9], Thm. 5.3.10), one obtains

$$\left.\frac{d}{d\tau_R} S(\rho\|\sigma_{\tau_R})\right|_{\tau_R=0} \;=\; -\langle\Omega_\rho|\, K_R\, |\Omega_\rho\rangle \;=\; -\langle K_R\rangle_\rho,$$

which is (5). The conversion to $S_{\rm gen}$ through the FS24 identification gives (6); the boundary terms vanish for the KMS reference σ on a Killing horizon (FS24, Sec. 3). □

Remark 1. *The regularity hypothesis $\rho \in \operatorname{dom} K_R$ is non-trivial in type-II$_\infty$: it requires ρ to lie in the form-domain of K_R, equivalent to a finite* modular energy *expectation. For Boulware or perturbed-thermal reference states in the SdS static patch, this condition is plausible but has not been verified state-by-state in the DEHK literature. We treat* (6) *as a conditional theorem in this sense.*

Remark 2 (Type-I sanity check)**.** *In the type-I limit (replace $\mathcal{M}_R$ by $\mathcal{B}(\mathcal{H})$ with $\sigma = \rho_\sigma \in \mathcal{S}_1(\mathcal{H})$ a density matrix), $\Delta_\sigma = \rho_\sigma \otimes \rho_\sigma^{-1}$ in the standard representation, $K_R = -\log\rho_\sigma$, and* (5) *reduces to the textbook formula $d/dt\, S(\rho\|\rho_\sigma e^{-tK_R})|_{t=0} = -\mathrm{Tr}(\rho K_R)$ of Uhlmann [11] and Lindblad [12]. The type-II$_\infty$ statement is the natural extension to a setting where ρ_σ is not trace-class.*

Compatibility with the FS24 GSL. Equation (6) is consistent with the Faulkner–Speranza GSL [4]: monotonicity of relative entropy under the modular inclusions of $\mathcal{M}_R$ along the horizon implies $d/d\tau_R\, S(\rho\|\sigma) \geq 0$ for cuts moving into the future (modular flow as "algebraic time"), and Theorem 1 identifies the rate with $\langle K_R\rangle_\rho$. This is the algebraic analogue of Wall's GSL theorem [13] as recast in the crossed-product language.

3 Wk3: Modular-rigidified Stinespring embedding

We now turn to the question whether K_R admits a holographic dual identification with the chord Hamiltonian of the DSSYK / sine-dilaton holographic dictionary.

3.1 The chord-side data (HPS / HOPSW)

HPS [5] construct, from the chord Hilbert space $\mathcal{H}_{\rm HPS} = \ell^2(\mathbb{N}_0)$ with chord-creation operator $a^\dagger$ and chord-number basis $|n\rangle = (a^\dagger)^n|0\rangle/\sqrt{[n]_q!}$, the gravitational transfer-matrix Hamiltonian

$$H_{\rm grav} \;=\; \frac{1}{\sqrt{2|\log q|(1-q^2)}}\Big[-\cos(\hat{P}) + \tfrac{1}{2}e^{i\hat{P}}e^{-\hat{L}}\Big], \tag{7}$$

acting on $\mathcal{H}_{\rm HPS}$, where $\hat{L} = 2|\log q|\hat{n}$ is the chord-length operator and $\hat{P}$ its conjugate. The spectrum is

$$\mathrm{spec}(H_{\rm grav}) \;=\; \Big\{\, E(\theta) = -\cos\theta\,\big[2|\log q|(1-q^2)\big]^{-1/2} : \theta \in [0,\pi] \,\Big\}, \tag{8}$$

which is bounded continuous on $[-E_{\max}, +E_{\max}]$ with $E_{\max} = [2|\log q|(1-q^2)]^{-1/2}$. The Plancherel measure is $d\mu(\theta) = (q^2, e^{\pm 2i\theta}; q^2)_\infty \, d\theta$ (HPS, eq. 28).

HOPSW [6] establish the holographic dictionary $\hat{L} \leftrightarrow$ extremal timelike volume in the dS_2 geometry anchored at past/future infinities, and identify $H_{\rm grav}$ with the ADM Hamiltonian of sine-dilaton dS_2 in the high-energy $\theta \to \pi$ limit. The Hawking temperature reads $\beta_{\rm dS}(\theta) = -2\pi/\theta$ in the dS regime (HOPSW, eq. (17) and following).

3.2 The crossed-product side (CLPW / DEHK)

On the gravity side, CLPW [2] and DEHK [3] construct the crossed-product algebra $\mathcal{M}_R = \mathcal{A}_R \rtimes_{\sigma^\phi} \mathbb{R}$ acting on $\mathcal{H}_{\rm DEHK} = \mathcal{H}_{\rm qft} \otimes L^2(\mathbb{R})$, generated by $\mathcal{A}_R$ (the bulk QFT type-III_1 algebra in the static patch), the dressed observables $e^{ipH_{\rm mod}}\, a\, e^{-ipH_{\rm mod}}$, and the clock momentum p. The dressed modular Hamiltonian is (CLPW, eq. (2.10) footnote; DEHK, eq. (3.6))

$$K_R \;=\; \beta_{\rm dS}\, H_\xi \;+\; p, \tag{9}$$

where H_ξ is the boost generator (continuous unbounded spectrum on $\mathbb{R}$ since $\mathcal{A}_R$ is type-III_1) and p is the clock momentum (continuous spectrum on $\mathbb{R}$ for an ideal clock). Hence

$$\mathrm{spec}(K_R) \;=\; \mathbb{R}, \tag{10}$$

unbounded both above and below.

3.3 The modular-rigidified Stinespring embedding C_{BS}

The natural attempt to bridge the two sides is a CPTP isometry $\Phi : \mathcal{B}(\mathcal{H}_{\rm HPS}) \to \mathcal{M}_R$ obtained from a Stinespring isometry $V : \mathcal{H}_{\rm HPS} \to \mathcal{H}_{\rm DEHK}$ with $V|n\rangle = |\xi_n\rangle$, where $\{|\xi_n\rangle\}_{n\geq 0}$ are analytic vectors in the natural positive cone $\mathcal{P}^\sharp$ of $(\mathcal{M}_R, \sigma)$ chosen so as to be eigenstates of $\hat{L}$ under the HOPSW dictionary. Berry-phase ambiguity in V is rigidified by the Caputa–Magán–Patramanis–Tonni [16] modular-Lyapunov universality $\lambda_L^{(\rm mod)} = 2\pi$ together with the Connes cocycle compatibility, fixing V up to a global phase.

The bridge would qualify as a *genuine* algebraic identification if it satisfied:

(P1) (state matching) $\Phi(\rho_{\rm HPS}) = \rho_{\rm DEHK}$,

(P2) (modular-flow intertwining) $\Phi \circ \sigma_t^{(\rm HPS)} = \sigma_t^{(\rm DEHK)} \circ \Phi$ for all $t \in \mathbb{R}$,

(P3) (Connes-cocycle equivariance) the cocycle $[D\rho_{\rm DEHK} : D\sigma_{\rm DEHK}]_t \in \mathcal{M}_R$ pulls back to a corresponding cocycle on $\mathcal{B}(\mathcal{H}_{\rm HPS})$.

The detailed construction of V, the Stinespring map $\Phi(B) = \pi_R(VBV^*)$, and the Berry-rigidification by the cocycle identity are reviewed in the predecessor memo [21]. What concerns us here is whether (P1)–(P3) are operator-level attainable.

4 Wk4: Spectrum-mismatch no-go

Proposition 1 (No-go for HPS–DEHK Stinespring intertwining)**.** *Let H_{grav} be the chord (transfer-matrix) Hamiltonian on $\mathcal{H}_{\mathrm{HPS}}$ with spectrum* (8) *bounded on $[-E_{\max}, +E_{\max}]$, and let K_R be the dressed modular Hamiltonian on $\mathcal{H}_{\mathrm{DEHK}}$ with spectrum $\mathbb{R}$. Then no isometry $V : \mathcal{H}_{\mathrm{HPS}} \to \mathcal{H}_{\mathrm{DEHK}}$ can intertwine H_{grav} with K_R in the sense of $V\,H_{\mathrm{grav}} = K_R\,V$ on $\operatorname{dom} H_{\mathrm{grav}}$. Consequently, the modular-rigidified Stinespring embedding $\Phi : \mathcal{B}(\mathcal{H}_{\mathrm{HPS}}) \to \mathcal{M}_R$ of Section 3 cannot satisfy (P2) in any non-trivial sense, and (P1), (P3) fail accordingly.*

Proof. Suppose, for contradiction, that $V\,H_{\mathrm{grav}} = K_R\,V$ on a core for H_{grav}. Since V is an isometry, VV^* is the orthogonal projection onto $\operatorname{ran} V \subset \mathcal{H}_{\mathrm{DEHK}}$, and $VH_{\mathrm{grav}}V^* = K_R\,VV^*$. Then for any unit vector $\psi \in \operatorname{dom} H_{\mathrm{grav}}$,

$$\langle V\psi|\,K_R\,|V\psi\rangle \;=\; \langle\psi|\,H_{\mathrm{grav}}\,|\psi\rangle \;\in\; [-E_{\max}, +E_{\max}]$$

by (8). Hence $\operatorname{ran} V$ would lie in the spectral subspace $\chi_{[-E_{\max},+E_{\max}]}(K_R)\,\mathcal{H}_{\mathrm{DEHK}}$, i.e. K_R restricted to $\operatorname{ran} V$ would be bounded. But $\operatorname{ran} V$ is infinite-dimensional (since V is an isometry from $\ell^2(\mathbb{N}_0)$), and the restriction of a self-adjoint operator with continuous spectrum on $\mathbb{R}$ to any infinite-dimensional invariant subspace $\mathcal{K}$ either inherits unbounded spectrum (if $\mathcal{K}$ is not contained in a finite-spectral band) or is unitarily equivalent to a direct sum of bounded operators on spectral subspaces (if $\mathcal{K}$ *is* contained in a bounded spectral band of K_R). In the second case, $\operatorname{ran} V$ lives in $\chi_{[-E_{\max},+E_{\max}]}(K_R)\,\mathcal{H}_{\mathrm{DEHK}}$, which is proper and inconsistent with V being an isometry whose image is modular-flow invariant under $\sigma_t^{(\mathrm{DEHK})} = e^{itK_R} \cdot e^{-itK_R}$ (since unbounded K_R generates a flow that escapes any bounded spectral band on a generic $\operatorname{ran} V$ direction). The contradiction forces (P2) to fail.

Bounded-band spectral sub-case (case 2a/2b refinement). The spectral measure μ_K of the modular Hamiltonian K_R on a bounded band $[-\Lambda, +\Lambda]$ is supported on a compact set; the modular flow $\sigma_t = e^{itK_R}$ acts unitarily and the orbit of any state $|\psi\rangle$ with $\langle\psi|K_R^2|\psi\rangle < \infty$ remains within the bounded band. By the Stone–von Neumann theorem applied to the bounded band, the Stinespring isometry V either factors through the bounded spectral projection (case 2a) or escapes to higher modes (case 2b). In case 2a, the isometry is bounded and the no-go fails to apply on that subspace, but the full image $\operatorname{ran} V$ must be infinite-dimensional (since V is an isometry from $\ell^2(\mathbb{N}_0)$), and at most finitely many modes can reside in the bounded-band projection, so case 2a cannot hold globally. In case 2b, the modular flow on $V|\psi\rangle$ has support outside any chosen bounded band, contradicting the bounded-band assumption on H_{grav}. Either case yields a contradiction, confirming that no such V can exist.

(P1) fails because the β_{dS}-KMS state for K_R in $\mathcal{M}_R$ is supported on the full real spectrum, while $\rho_{\mathrm{HPS}} = \sum_n p_n(\beta)|n\rangle\langle n|$ has support in the discrete chord basis with thermal weights determined by E_n^{chord} from the bounded spectrum (8); $\Phi(\rho_{\mathrm{HPS}})$ then has bounded modular energy support $[-E_{\max}, +E_{\max}]$ inside $\mathcal{M}_R$, while the genuine $\rho_{\mathrm{DEHK}} = e^{-\beta_{\mathrm{dS}}K_R}/Z_{\mathrm{DEHK}}$ does not. (P3) is a corollary, since the Connes cocycle is built from the relative modular operator $\Delta_{\rho_{\mathrm{DEHK}}|\sigma_{\mathrm{DEHK}}}$, which in turn requires (P1). □

Remark 3 (Categorical character of the obstruction)**.** *The obstruction is not a small-parameter / loop-correction artifact: it is* topological *on the spectrum. Even with arbitrarily small $|\log q|$, $E_{\max} \to \infty$ but $E(\theta)$ remains bounded for each fixed $q \in (0,1)$, while K_R is unbounded on $\mathbb{R}$ for every $\beta_{\mathrm{dS}} > 0$. The two spectra never agree as sets.*

Remark 4 (On the type-I vs type-II$_\infty$ algebra mismatch)**.** *A coarser way to state the obstruction is that $\mathcal{B}(\mathcal{H}_{\mathrm{HPS}})$ is type-I (the chord algebra is finite-dimensional at any chord-truncation level N_{chord} and infinite-dimensional but separable in the limit; its von Neumann completion is $\mathcal{B}(\mathcal{H}_{\mathrm{HPS}})$, type-I), while $\mathcal{M}_R$ is type-II$_\infty$. Type-changing CPTP maps between general von Neumann algebras do exist (Faulkner–Hollands II [19]) and even admit FH-II approximate recovery channels under small relative-entropy perturbations. What Proposition 1 shows is that the* spectral obstruction *is sharper than the algebra-type obstruction: even within the type-changing FH-II framework, the candidate spectra $E(\theta)$ and* $\mathrm{spec}(K_R)$ *are incompatible.*

5 What survives: Krylov-thermodynamic match

The negative content of Section 4 is partial: while no operator-level isometry intertwines H_{grav} with K_R, the *thermodynamic* content of the two systems agrees at semiclassical order.

Proposition 2 (Krylov–thermodynamic match)**.** *In the regime $q \to 1^-$, $\theta \to \pi$ (the dS large-N regime of HPS–HOPSW), the partition functions $Z_{\mathrm{chord}}(\beta) := \int_0^\pi (q^2, e^{\pm 2i\theta}; q^2)_\infty \, e^{-\beta E(\theta)} \, d\theta$ and $Z_{\mathrm{DEHK}}(\beta) := \tau_{\mathrm{II}}(e^{-\beta K_R})$ satisfy, at the de Sitter inverse temperature β_{dS}:*

$$\log Z_{\mathrm{chord}}(\beta_{\mathrm{dS}}) \;=\; \log Z_{\mathrm{DEHK}}(\beta_{\mathrm{dS}}) \;+\; O(1/N), \tag{11}$$

both reproducing the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A_{\mathrm{horizon}}/(4G_N)$ at leading order.

Sketch. This is essentially a re-statement of (i) the standard Bekenstein–Hawking entropy of the cosmological horizon in CLPW/DEHK at leading $1/G_N$ [2, 3], and (ii) HOPSW eq. (27) (chord-Krylov disk partition function $\to$ dS$_2$ Bekenstein–Hawking entropy) [6], both regulated through the same β_{dS}-KMS thermal data on the cosmological horizon. At higher orders, the $1/N$ corrections differ: chord-side corrections follow from the q-Hermite Plancherel measure expansion; DEHK-side corrections follow from the loop expansion around the saddle of $\tau_{\mathrm{II}}(e^{-\beta K_R})$. We do not claim agreement beyond leading order. □

Remark 5. *Proposition 2 is* not *a new result: the Bekenstein–Hawking entropy of the cosmological horizon is recovered by* every *consistent dS holography proposal at semiclassical order. We include it here only to record* exactly *what part of the attempted bridge survives the no-go in Proposition 1: the partition-function thermodynamics, but not the underlying operator content.*

Remark 6 (Krylov-asymptotic match)**.** *A second weaker statement that survives is a Lanczos-asymptotic match. The HPS chord algebra has Lanczos coefficients $b_n = \sqrt{[n]_q(1-q^n)} \sim \alpha n + \gamma$ at large n (linear growth, signature of $\lambda_L = 2\pi T$). For K_R, Vardian's modular Krylov procedure [17] (in an AdS/CFT setting; transfer to dS is by methodological analogy) yields modular Lanczos coefficients b_n^{mod} also growing linearly at late modular time with $\lambda_L^{(\mathrm{mod})} = 2\pi$, the Caputa–Magán–Patramanis–Tonni universality [16]. Both spectral functions exhibit the same asymptotic Lyapunov exponent, but as different Jacobi matrices with different short-distance / low-n structure. This is a Krylov-asymptotic match, not an operator match, and is consistent with Proposition 1.*

6 Discussion and open problems

What is genuinely new. Theorem 1 is, as far as we are aware, the first explicit statement of the Araki-cocycle identity $dS_{\mathrm{gen}}/d\tau_R = \langle K_R \rangle_\rho$ in the type-II$_\infty$ crossed-product setting of CLPW/DEHK. It complements the FS24 GSL [4] by giving an explicit *rate* formula in terms of the modular-energy expectation, and makes the modular-time first law $\delta S_{\mathrm{gen}} = \beta_{\mathrm{dS}}\, \delta\langle K_R \rangle$ algebraically explicit on the type-II$_\infty$ trace.

What is genuinely refuted. The natural attempt to identify $\langle K_R \rangle_\rho$ with chord-Krylov complexity C_K via a Stinespring isometry between the DSSYK chord Hilbert space and the GNS Hilbert space of $\mathcal{M}_R$ is refuted at the operator level by Proposition 1, because the spectra of the chord Hamiltonian and K_R have incompatible support (bounded vs unbounded). What survives is the thermodynamic content of Proposition 2, i.e. the standard Bekenstein–Hawking dS entropy match, and the Krylov-asymptotic universal $\lambda_L = 2\pi$ shared between the two Lanczos sequences.

Alternative routes. Three logically distinct alternatives to the Stinespring intertwining route remain open:

1. *Type-changing FH-II recovery.* Faulkner–Hollands II [19] (and the predecessor FHSW [18]) construct universal approximate recovery channels for 2-positive maps between general von Neumann algebras, including the type-changing case. Their bound $S(\rho\|\sigma) - S(\Phi(\rho)\|\Phi(\sigma)) \geq -2\log F(\rho, R\circ\Phi(\rho))$ controls relative-entropy loss without requiring spectral matching of the modular Hamiltonians. A reformulation of the bridge problem in FH-II language might give an *approximate-recovery* statement that escapes the spectrum-mismatch no-go, but at the cost of giving up exact intertwining.

2. *Bounded function of K_R.* A bounded function $F(K_R)$ (e.g. $F(K_R) = (2/\pi)\arctan(\beta_{\mathrm{dS}} K_R/2)$) has spectrum in $(-1, +1)$ and could in principle match a rescaled H_{grav}. This is an ad hoc construction without independent physical motivation; we mention it for completeness only.

3. *Modular density-of-states matching.* A genuine result at the level of the spectral measure of K_R, namely that the chord-Krylov density of states $\rho(\theta)$ matches a suitable projection of the spectral measure of K_R in a controlled high-energy / late-modular-time limit, would yield a v6.2-style consistency note rather than a theorem. This route uses the CMPT [16] modular-Lyapunov universality and the modular Krylov approach of Vardian [17] (with the caveat that Vardian works in AdS/CFT and the dS extension requires an independent construction).

Scope of Vardian (2026) and the dS analogy. Vardian [17] (arXiv:2602.02675, live-verified) develops modular Krylov complexity as a boundary probe of the area operator and entanglement islands in AdS/CFT. That construction is set entirely within the holographic AdS/CFT framework, using the quantum extremal surface structure and operator-algebra quantum error correction. The de Sitter extension proposed in the present paper—using modular Krylov techniques in the crossed-product algebra of the static patch—is a *methodological analogy*, not a theorem transfer: the QES island construction and the OAQEC dictionary of Vardian do not have off-the-shelf de Sitter

counterparts. We use Vardian's work only as motivation for the bounded-band sub-case analysis in Proposition 1 and the Krylov-asymptotic match of §5; no specific theorem from Vardian is invoked in our proofs.

Closing assessment. The ER=EPR programme in the algebraic crossed-product setting yields, at the rigorous mathematical level, the Araki-cocycle rate formula (6) as a genuine new theorem (Theorem 1), and a clean operator-level no-go for the chord–K_R bridge (Proposition 1). The survival is the standard Bekenstein–Hawking thermodynamic match (Proposition 2), which was already known from independent considerations.

We do not claim that the ER=EPR / chord-vs-modular-complexity programme is closed, only that the most natural "Stinespring isometry" bridge is ruled out, and that future work must use either type-changing approximate recovery in the FH-II sense, a bounded-functional-calculus construction, or a coarser density-of-states / Lanczos-asymptotic matching at the level of spectral measures rather than operator identifications.

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